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ENGINEERING AND TECHNOLOGY,
MYSORE**

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VTU Previous Question Papers

“Electronics and Communication Engineering”

**IV, VI & VIII Semester
Dec.2024/Jan.2025**

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Bannur road, Alanahalli, Mysore-570028

INDEX

S/L	Subject Code	Subject Title	Exam Date
1	BBOK407	Biology for Engineers	Dec.2024/Jan.2025
2	BEC401	Electromagnetic Theory	Dec.2024/Jan.2025
3	BEC402	Principles of Communication Systems	Dec.2024/Jan.2025
4	BEC403	Control Systems	Dec.2024/Jan.2025
5	BUHK408	Universal Human Values	Dec.2024/Jan.2025
6	21EC/EI61	Technological Innovation Management and Entrepreneurship	Dec.2024/Jan.2025
7	21EC62	Microwave Theory and Antennas	Dec.2024/Jan.2025
8	21EC63	VLSI Design and Testing	Dec.2024/Jan.2025
9	21EC641	Artificial Neural Networks	Dec.2024/Jan.2025
10	21EC643	Python Programming	Dec.2024/Jan.2025
11	21MAT655	Advanced Linear Algebra	Dec.2024/Jan.2025
12	18EC42	Analog Circuits	Dec.2024/Jan.2025
13	18EC43	Control Systems	Dec.2024/Jan.2025
14	18EC44	Engineering Statistics and Linear Algebra	Dec.2024/Jan.2025
15	18EC45	Signals and Systems	Dec.2024/Jan.2025
16	18EC61	Digital Communication	Dec.2024/Jan.2025
17	18EC62	Embedded Systems	Dec.2024/Jan.2025
18	18EC63	Microwave and Antenna	Dec.2024/Jan.2025
19	18EC642	Artificial Neural Networks	Dec.2024/Jan.2025
20	18EC643	Data Structure using C++	Dec.2024/Jan.2025
21	18MAT41	Complex Analysis, Probability and Statistical Methods	Dec.2024/Jan.2025

CBCS SCHEME

USN

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BBOK407

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Biology for Engineers

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Define cell. Explain the structure and function of plant cell with neat diagram.	08	L2	CO1
	b.	Define Stem Cell. Discuss the types and application of stem cells.	06	L2	CO1
	c.	Describe the properties and functions of hormones.	06	L2	CO1
OR					
Q.2	a.	Discuss the properties and functions of nucleic acids in cellular processes.	07	L2	CO1
	b.	Discuss the properties and functions of enzymes.	07	L2	CO1
	c.	Discuss the properties of vitamins and its supplies.	06	L2	CO1
Module – 2					
Q.3	a.	Apply the knowledge of nucleic acid in DNA finger printing in forensic applications.	08	L3	CO1
	b.	Discuss whey protein and plant based protein as protein based food.	06	L2	CO1
	c.	Write a note on PLA as bioplastic.	06	L1	CO1
OR					
Q.4	a.	Apply your knowledge of lipids and outline the process of obtaining biodiesel from lipids.	07	L3	CO1
	b.	Define vaccine. Discuss the mechanism of RNA vaccine for COVID-19.	07	L2	CO1
	c.	Write a note on enzyme based biosensors.	06	L1	CO1
Module – 3					
Q.5	a.	Compare human brain with computer's CPU.	07	L3	CO2
	b.	Explain lungs as a purification system.	07	L2	CO2
	c.	Write a note on dialysis systems of kidney.	06	L1	CO2
OR					
1 of 2					

Q.6	a.	Illustrate the engineering solutions available for Parkinson's disease.	07	L3	CO2
	b.	Explain heart as a pumping system.	07	L2	CO2
	c.	Write a note on optical correction and materials used for lens.	06	L1	CO2
Module – 4					
Q.7	a.	Illustrate the HBOCs and PFCs as human substituents.	07	L3	CO3
	b.	Explain how the structure of shark skin reduces drag and how these properties have been applied to improve swim suit.	07	L2	CO3
	c.	Explain the term GPS and aircrafts technology as bio inspired by bird fly.	06	L2	CO3
OR					
Q.8	a.	Compare the uses of ultrasonography and sonars.	07	L3	CO3
	b.	Discuss the king fisher beak shaped bullet train to the reduction of noise and improve the stability.	06	L2	CO3
	c.	Explain the term superhydrophobic and self-cleaning in lotus leaf effect.	07	L2	CO3
Module – 5					
Q.9	a.	Explain bioimaging and artificial intelligence technique in disease diagnosis.	08	L2	CO4
	b.	Explain the working principles of electrical tongue and electrical nose in food industry.	06	L2	CO4
	c.	Write a note on bioengineering of Muscular dystrophy and osteoporosis.	06	L1	CO4
OR					
Q.10	a.	Explain the process of biomining via microbial surface adsorption.	07	L2	CO4
	b.	Describe the concept of DNA origami and its role in bio-computing.	06	L2	CO4
	c.	Write a note on self healing bio-concrete and bio-mineralization processes.	07	L1	CO4

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Electromagnetic Theory

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	State and explain Coulomb's law of force between two point charges in vector form.	8	L1	CO1
	b.	Define Electric field intensity. Derive the expression for the electric field intensity at a point due to infinite line charges (Uniformly charged wire).	8	L2	CO1
	c.	Two very small conducting spheres, each of mass 1×10^{-4} kg are suspended at common point by very thin filaments of length 0.2m. A charge Q Coulomb is placed on each sphere. The electric force of repulsion separates the spheres and an equilibrium is reached when the suspending filaments make an angle of 10° . Assuming $e_r = 1$, $g = 9.8\text{N/kg}$ and negligible mass for the filaments, find Q.	4	L3	CO1
OR					
Q.2	a.	Define Point charge and using Coulomb's Law, derive expression for electric field intensity due to a point charge.	8	L2	CO1
	b.	Let a point $Q_1 = 25\text{nc}$ be located at A(4, -2, 7) and a charge $Q_2 = 60\text{nc}$ be at B(-3, 4, -2). Find $\vec{E}$ at C(1, 2, 3). Also find direction of the electric field. Given $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$.	8	L3	CO1
	c.	Two point charges of $+3 \times 10^{-9} \text{ C}$ and $-2 \times 10^{-9} \text{ C}$ are spaced two meter apart. Determine the electric field at a point which is one meter from each of the two point charges.	4	L3	CO1
Module – 2					
Q.3	a.	State and prove Gauss Divergence theorem or divergence theorem.	8	L2	CO2
	b.	A point charge, $Q = 30\text{nc}$ is located at the origin in Cartesian coordinates. Find the electric flux density and electric field intensity at (1, 3, -4)m.	8	L3	CO2
	c.	Derive an equation for equation of continuity (continuity of current).	4	L3	CO2
OR					
Q.4	a.	State and prove Gauss law.	8	L2	CO2
	b.	Given that the potential field is $V = 2x^2y - 5z$. Find the potential, electric field intensity and volume charge density at point P(-4, 3, 6).	8	L3	CO2
	c.	State Gauss law in point form. Hence derive Maxwell's first equation.	4	L3	CO2

Module – 3					
Q.5	a.	Starting from gauss law, derive Poisson's and Laplace equation. Hence define Laplace equation in all three coordinate systems.	4	L2	CO3
	b.	State and prove Stoke's theorem.	8	L2	CO3
	c.	Find the potential and volume charge density at P(0.5 , 1.5 , 1)m in free space. Given the potential field as under. i) $V = 2x^2 - y^2 - z^2$ volt ii) $V = 6 r \phi z$ volt.	8	L3	CO3
OR					
Q.6	a.	State and prove Biot – Savart's law.	4	L1	CO3
	b.	State and prove Ampere's circuital law.	8	L1	CO3
	c.	The magnetic field intensity is given in a certain region of space as : $\vec{H} = \left(\frac{x+2y}{z^2} \right) \hat{a}_y + \frac{2}{z} \hat{a}_z$ A/m. i) Find $\nabla \times \vec{H}$ ii) Find $\vec{J}$ iii) Use $\vec{J}$ to find total current passing through the surface , $Z = 4$, $1 < x < 2$, $3 < y < 5$ in the $\hat{a}_z$ direction.	8	L3	CO3
Module – 4					
Q.7	a.	Define current element. Derive an equation for force on a differential current element in a magnetic field.	8	L2	CO4
	b.	A point charge $Q = 18\text{nc}$ has a velocity of 5×10^6 m/s in the direction $\vec{a} = 0.6 \hat{a}_x + 0.75 \hat{a}_y + 0.3 \hat{a}_z$. Calculate the magnitude of the force exerted on the charge by the field $\vec{B} = -3 \hat{a}_x + 4 \hat{a}_y + 6 \hat{a}_z$ mT.	8	L3	CO4
	c.	Calculate the force on a straight conductor of length 0.3m carrying a current 5A in the Z – direction where the magnetic field is $\vec{B} = 3.5 \times 10^{-3} (a\hat{x} - a\hat{y})$ Tesla. ($a\hat{x}$ and $a\hat{y}$ are unit vectors).	4	L3	CO4
OR					
Q.8	a.	Derive magnetic boundary condition for i) Tangential component of magnetic field. ii) Normal component of magnetic field.	8	L2	CO4
	b.	A conductor 4m long lies along the Y – axis with a current of 10A in the $a\hat{y}$ direction. Find the force on the conductor if the field in the region is $\vec{B} = 0.05 a\hat{x}$ tesla.	8	L3	CO4
	c.	Find the magnetic field intensity inside a magnetic material for following conditions : $M = 100\text{A/m}$ and $\mu = 1.5 \times 10^{-5}$ H/m $B = 200\mu\text{T}$, X_m (Magnetic susceptibility = 15).	4	L3	CO4

Module – 5					
Q.9	a.	Derive Integral and point form of Faraday's law.	8	L2	CO5
	b.	Given $\vec{E} = E_m \sin (wt - \beta z) \hat{a}_y$ in free space. Calculate $\vec{D}$, $\vec{B}$ and $\vec{H}$.	8	L3	CO5
	c.	A copper disc 40cm diameter is rotated at 3000 r.p.m on a horizontal axis perpendicular to and through the centre of disc axis, lying in magnetic meridian. Two brushes make contact with the disc at diametrically opposite points on the edge. If horizontal component of earth's field is 0.02 mT, find the induced e.m.f between brushes.	4	L3	CO5
OR					
Q.10	a.	State and derive Poynting's theorem for uniform plane waves.	8	L2	CO5
	b.	Derive general wave equation in electric and magnetic fields.	8	L2	CO5
	c.	For silver, the conductivity is $\sigma = 3.0 \times 10^6$ s/m. At what frequency will depth of penetration be 1mm?	4	L3	CO5

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Principles of Communication Systems

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define probability. Illustrate the relationship between sample space, events and probability.	06	L1	CO1
	b.	Outline random processes and illustrate an ensemble of sample function with a neat diagram.	06	L2	CO1
	c.	Show that if a Gaussian process $x(t)$ is applied to a stable linear filter, then the random process $y(t)$ developed at the output of the filter is also Gaussian.	08	L3	CO2
OR					
Q.2	a.	What is conditional probability? Prove that $P(B/A) = P(A/B) \cdot P(B) / P(A)$	06	L1	CO1
	b.	Define mean, correlation and covariance function.	06	L2	CO2
	c.	Develop a program to generate the probability density function of Gaussian distribution function.	08	L3	CO2
Module – 2					
Q.3	a.	An antenna has an impedance of 40Ω an unmodulated AM signal produces a current of 4.8 A. The modulation is 90 percent calculate i) The carrier power ii) The total power iii) The sideband power	06	L1	CO1
	b.	Explain with neat diagrams amplitude demodulation using the diode detector.	07	L1	CO1
	c.	Explain a general block diagram of an FDM system	07	L2	CO2
OR					
Q.4	a.	Interpret the concept of modulation index and percentage of modulation write the necessary equations.	06	L1	CO1
	b.	Explain high level collector modulation with neat block diagram.	07	L2	CO1
	c.	Explain with diagrams the working principle of lattice type balanced modulator.	07	L2	CO2
Module – 3					
Q.5	a.	Compare and contrast FM and AM.	06	L1	CO1
	b.	Explain with diagrams the working principle of frequency modulation using voltage controlled oscillator.	07	L2	CO2
	c.	Explain general block diagram of a super heterodyne receiver.	07	L2	CO2
OR					
Q.6	a.	The input to an FM receiver having an S/N of 2.8. The modulating frequency is 1.5 KHz. The maximum permitted deviation is 4 KHz. What are (i) The frequency deviation caused by the noise (ii) The improved output S/N.	06	L2	CO2
	b.	Define PLL. Explain the basic block diagram of a PLL.	07	L1	CO2
	c.	Explain JFET mixer.	07	L2	CO2

Module – 4					
Q.7	a.	What are the advantages of digital signal over analog signals?	04	L1	CO1
	b.	Explain with basic elements of a PCM system with neat diagrams.	08	L2	CO1
	c.	For the data stream 0 1 1 0 1 0 0 1 draw the following line code waveforms i) Unipolar NRZ ii) Polar NRZ iii) Unipolar RZ iv) Manchester code	08	L3	CO2
OR					
Q.8	a.	State and prove Sampling theorem.	04	L1	CO1
	b.	What is multiplexing and why is it required in communication? Explain the working of TDM with a neat block diagram.	08	L2	CO1
	c.	Explain the generation of PPM with a relevant block diagrams and waveforms.	08	L2	CO2
Module – 5					
Q.9	a.	Define Intersymbol interference (ISI) outline baseband binary data transmission system with neat block diagram and equations.	08	L2	CO1
	b.	Develop a code to generate RZ pulse.	04	L3	CO2
	c.	Define signal to noise ratio. Explain different types of external and internal noise.	08	L2	CO1
OR					
Q.10	a.	Explain the following concept briefly: i) Nyquist criterion for distributors transmission ii) Baseband M-ary PAM transmission	08	L1	CO2
	b.	Develop a code to generate Raised cosine pulse.	04	L2	CO2
	c.	Illustrate the concept of noise in cascaded stages with a diagram. Write Friis formula and mention its terms.	08	L2	CO3

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Control Systems

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module - 1			M	L	C
Q.1	a.	Compare open loop and closed loop control system with practical example.	06	L2	CO1
	b.	For the system shown in Fig.Q1(b). Find the transfer function $G(s) = \frac{\theta_2(s)}{T(s)}$ consider $J_1 = 1 \text{ kgm}^2$, $K_1 = 1 \text{ Nm/rad}$, $K_2 = 1 \text{ Nm/rad}$, $B_1 = 1 \text{ Nm/rad/sec}$, $B_2 = 1 \text{ Nm/rad/sec}$.	06	L2	CO1
		Fig.Q1(b)			
	c.	Draw the mechanical network for the system shown in Fig.Q1(c). Write the equations of performance and draw its analogous circuit based one force voltage analogy.	08	L2	CO1
		Fig.Q1(c)			
OR					
Q.2	a.	The circuit shown in Fig.Q2(a) is called lead-lag filter. Find the transfer function $\frac{V_2(s)}{V_1(s)}$ when $R_1 = 100 \Omega$, $R_2 = 200 \text{ K}\Omega$, $C_1 = 1 \mu\text{F}$ and $C_2 = 0.1 \mu\text{F}$.	10	L3	CO1
		Fig.Q2(a)			

- b. What are the variables and elements of translational motion? For the mechanical system shown in Fig.Q2(b).
- Write the differential equations of performance.
 - Draw and write loop and nodal equations based on F-V and F-I analogous networks.

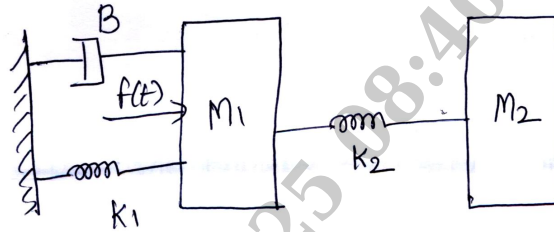


Fig.Q2(b)

Module – 2

- Q.3 a. Give any six block diagram reduction rules to find the transfer function of the system. **04** **L1** **CO2**

- b. For the system represented in the given Fig.Q3(b), determine transfer function $C(s)/R(s)$. **06** **L2** **CO1**

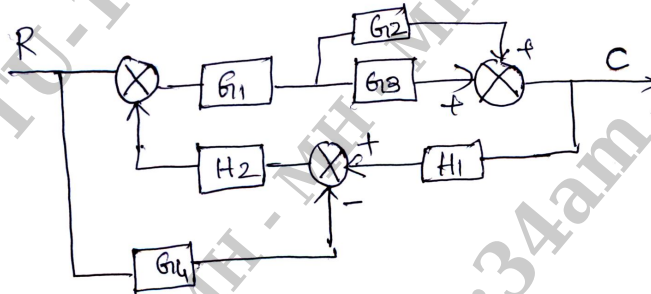


Fig.Q3(b)

- c. Find the overall transfer function of the system whose signal flow graph is shown in Fig.Q3(c). **10** **L2** **CO2**

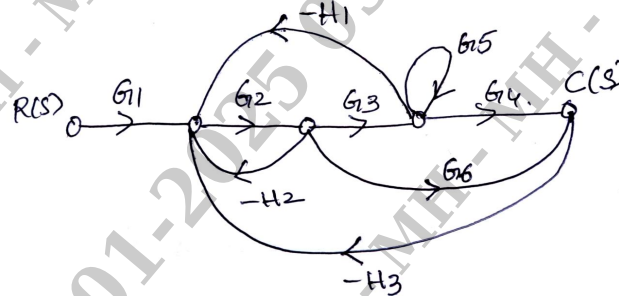


Fig.Q3(c)

OR

- Q.4 a. Interpret the transfer function by converting the block diagram into signal flow graph. **10** **L2** **CO2**

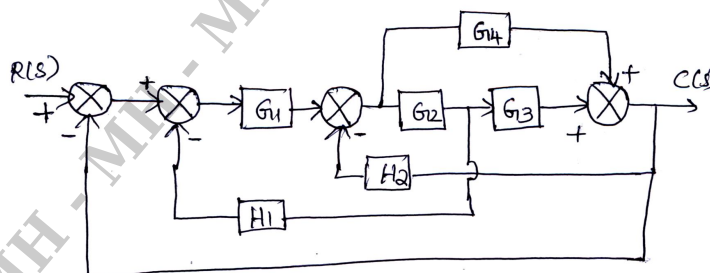


Fig.Q4(a)

- b. Obtain the transfer function for the block diagram shown in Fig.Q4(b) using block diagram reduction technique.

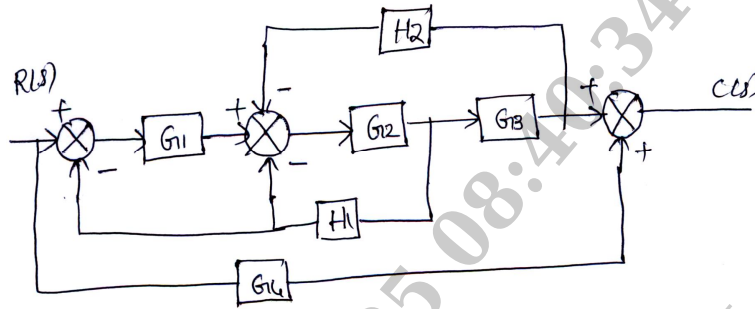


Fig.Q4(b)

Module – 3

- Q.5** a. Make use of the response curve of 2nd order under-damped system to define and derive the expression for (i) peak time (ii) peak overshoot (iii) rise time **10 L2 CO3**
- b. Find K_p , K_v and K_a for a system having $G(s) = \frac{s+10}{s(s^3+7s^2+12s)}$. Also, evaluate the steady state error, when the I/P $r(t)$ is given by:
 (i) $r(t) = 5u(t)$ (ii) $r(t) = 2t u(t)$ (iii) $r(t) = 4t^2 u(t)$ **10 L2 CO3**

OR

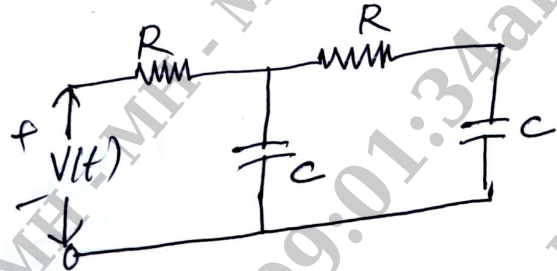
- Q.6** a. Derive an expression for the under damped response of a second order feedback control system for step input. **10 L2 CO2**
- b. Explain the static error constant and derive the expressions. **06 L2 CO2**
- c. Analyze the effect of PD controller for 2nd order control system with appropriate equations. **04 L2 CO2**

Module – 4

- Q.7** a. The open loop transfer function of a unity feedback system is given by $G(s) = \frac{K}{s(s+3)(s^2+s+1)}$. Find the value of K that will cause sustained oscillation and hence find the oscillation frequency. **08 L2 CO3**
- b. Sketch the root locus plot for a negative feedback control system whose open loop transfer function is given by $G(s)H(s) = \frac{K}{s(s+1)(s+2)(s+3)}$. For all values of K ranging from 0 to ∞ . Find the value of K for closed loop stability. **12 L3 CO3**

OR

- Q.8** a. For the characteristic equations given below, determine number of roots with positive real part:
 i) $s^6 + s^5 + 3s^4 + 2s^3 + 5s^2 + 3s + 1 = 0$
 ii) $s^8 + s^7 + 4s^6 + 3s^5 + 14s^4 + 11s^3 + 20s^2 + 9s + 9 = 0$ **10 L2 CO4**

	b.	Show that the part of root locus of a system with $G(s)H(s) = \frac{K(s+3)}{s(s+2)}$ is a circle having center $(-3, 0)$ and radius at $\sqrt{3}$.	10	L3	CO3
Module – 5					
Q.9	a.	Construct the bode plot for the transfer function $G(s) = \frac{80}{s(s+2)(s+20)}$. Determine GM and PM, ω_{pc} , ω_{gc} .	10	L2	CO3
	b.	Obtain the state transition matrix for the following system: $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -0.5 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} u$	10	L2	CO5
OR					
Q.10	a.	Using Nyquist stability criteria investigate the stability negative feedback control system whose open loop transfer function is given by $G(s)H(s) = \frac{100}{(s+1)(s+2)(s+3)}$. Assume $\omega_g = 1.253$ rad/sec.	10	L2	CO5
	b.	Obtain the state model of electrical network shown in Fig.Q10(b), by choosing $V_1(t)$ and $V_2(t)$ as state variables.  Fig.Q10(b)	10	L3	CO5

CBCS SCHEME

BUHK408

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Question Paper Version : A

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Universal Human Values

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

1. What is the state of liking and a holistic and all-encompassing state of the mind that crates inner harmony?
a) Prosperity b) Happiness c) Innateness d) Organized
2. _____ is defined as a group of people working together to create a surplus
a) Alone b) Organisation c) System d) Army
3. The process whereby national economies and business systems are becoming deeply interlinked with each other is called
a) Nationalization b) Glocalization c) Global linking d) Globalization
4. Which of the following actions will likely lead to organizational ethical behavior?
a) Establishing an ethics office
b) Promoting moral courage
c) Developing strong governance processes
d) All of these
5. _____ behavior tends to arise when managers decide to put the attainment of their own personal goals, or the goals of the organization, above the fundamental rights of one or more stakeholder groups
a) Complementary b) Situational c) Unethical d) Confusing
6. Accepted principles of right or wrong governing the conduct of business people are called :
Choose one answer
a) Business b) Conduct c) Business ethics d) Principles
7. What is true about a professional
a) The professional has to have autonomy
b) He cannot be supervised by the client
c) He should not subject to political control
d) All of these

8. The basic rule of professionalism is “not knowingly to do _____”
a) Harm b) Work c) Competition d) Copying
9. The study of _____, as contrasted with legal, obligations is called ethics
a) Behaviors b) Moral c) Business d) All of these
10. Ethics is a conception of _____ conduct
a) Bad/Wrong b) Good/Correct c) Right/True d) Right and Wrong
11. Ethics and _____ go together
a) Profits b) Management c) Worker ship d) All of these
12. _____ driven companies are always successful in the long run
a) Management b) Worker union c) Value d) Profit
13. A business needs to be ethical, in order to
a) Fulfill public expectations for business
b) Prevent harming others
c) Improve business relations and employees productivity
d) All of these
14. Following is (are) types(s) of ethics issues(s)
a) Functional area ethics b) Corporate policy ethics
c) Face to face ethics d) All of these
15. What is called living with assumption for oneself as body and living of human being only on the basis of physical facilities, and not with right understanding and relationship?
a) Human consciousness b) Happiness
c) Right Understanding d) Animal Consciousness
16. Samridhi means
a) Happiness b) Wealth c) Prosperity d) Health
17. Developed nations are the live example of
a) Prosperity b) Wealth c) Happiness d) Health
18. The participation of human beings is seen in two forms
a) Prosperity and work b) Values and Understanding
c) Behavior and Wealth d) Behavior and Work
19. What are the outcomes of realization and understanding?
a) Work b) Values c) Happiness d) Health
20. Corporate culture _____ refers to how widely and deeply employees hold the company’s dominant values and assumptions
a) Breadth b) Depth c) Character d) Strength
21. The observable symbols and signs of a organization’s culture are called :
a) Its cultural design b) Artifacts c) Cultural formatting d) Its cultural dcor
22. _____ is the feeling or responsibility and commitment for ensuring Right Understanding and Right Feeling in the self of my relative
a) Guidance b) Care c) Respect d) Affection
23. The ability to perform a specific task, action, or function successfully is called _____
a) Competence b) Purpose c) Intention d) Program

24. To live in harmony with other human beings is ____
 a) Svatva b) Swatantrata c) Swarajya d) None of these
25. Natural acceptance is?
 a) Universal b) Definite c) Continuous d) All of these
26. Basic human aspirations are ____
 a) Happiness b) Prosperity c) Education d) Both (a) and (b)
27. Which of the following is true?
 a) Happiness is definite and it can be understood
 b) I need to be unhappy to recognize that I am happy
 c) One will be bored of happiness if one is always happy
 d) I will grow only if I am unhappy. If I become happy my growth will stop
28. To be wealthy is ____ condition in the modern world to be happy
 a) Not essential b) True c) Fake d) The most important
29. It is important to have a good ____ to be successful in your career
 a) Appear b) Attend c) Attitude d) Respectful
30. ____ skills such as writing, speaking and body language are important work ethics so that you can be clear in your messages to others
 a) Communication b) Cooperation c) Honesty d) Integrity
31. You are a person with a high level of ____ if you stand by you stand by your values even if others put pressure on you to do something that is wrong or if they make fun of you
 a) Communication b) Cooperation c) Honesty d) Integrity
32. If you don't have ____ you can really get in trouble for lying, cheating or deceiving others
 a) Communication b) Cooperation c) Honesty d) Integrity
33. If you have ____ you will be a trusted person because they will see that you are committed to your company
 a) Loyalty b) Organizational Skills c) Productivity d) Respect
34. Some jobs cannot be done alone and some can be done much better and faster if you possess the quality of ____
 a) Responsibility b) Respect c) Teamwork d) Cooperation
35. The three major types of ethical issues include except?
 a) Communication issues b) Systematic issues
 c) Corporate issues d) Individual issues
36. Ethics are moral principles and values which ____
 a) Guide a firm's behavior
 b) Govern the actions of an individual
 c) Provide employees with rules on how to behave
 d) Are legally enforceable
37. The word "Ethics" comes from
 a) Ethos b) Eternity c) Elementary d) Essentiality
38. A ____ is a set of principles that people hold themselves to or that are held by organizations or groups
 a) Code of ethics b) Code of honor c) Ethical perspective d) Persuasive tactic

39. _____ are beliefs and attitudes we have that can actually conflict with our ethical decision
a) Morals b) Principles c) Virtues d) Values
40. Intellectual property rights are an important aspect in international trade. Which of the following is covered under IPR?
a) Patents b) Copyrights c) Trademarks d) All of these
41. Which of the following is not the infringement of copyright?
a) Copy the software to another computer by educational institution
b) Copy the software to another computer by company
c) Copy the software to another computer
d) To make backup copies
42. Patent can be granted for _____?
a) Process b) Ideas
c) Both product and process d) Product
43. Government may order the non advertisement of any patent application in the case of invention related to _____
a) Information technology b) Food
c) Drug d) Defense technology
44. Patent for the _____ is prohibited
a) Traditional knowledge b) Arms
c) Atomic energy d) Junk food
45. A design can be also registered under _____
a) Trademark Act b) Patent Act
c) Geographical Indication Act d) Copyright Act
46. IPR Complete specification gives _____
a) Full description of claim only
b) Full description of royalty
c) Full description of invention
d) Full description of application
47. Is the first owner of a Book's copyright?
a) Author b) Publisher c) Translator d) Printing press owner
48. Copyright is a _____
a) Creation by person of efforts, intellectual and capital
b) Creation by person of the efforts only
c) Creation by negligence of a person
d) Creation by the mental status of a person
49. Globalization has improved the living structure of which of the following?
a) All the people
b) People living in developing countries
c) People living in developed countries
d) None of these
50. Which sector experienced the least profit due to the effect of Globalization?
a) Industrial sector b) Service sector c) Agriculture Sector d) Secondary sector

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Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Technological Innovation Management and
Entrepreneurship

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Management and explain the functions of management. (08 Marks)
- b. Is management a science or art? Explain. (06 Marks)
- c. Explain the various roles of manager. (06 Marks)

OR

- 2 a. Explain the different levels of management along with skills required. (07 Marks)
- b. Distinguish between Management and Administration. (05 Marks)
- c. Explain the various steps involved in decision making process. (08 Marks)

Module-2

- 3 a. Briefly explain the principles of organization. (10 Marks)
- b. What is Departmentalisation? Explain the advantages and disadvantages of committees in an organization. (10 Marks)

OR

- 4 a. Explain Maslow's hierarchy need theory. (10 Marks)
- b. Explain the various steps involved in selection process. (10 Marks)

Module-3

- 5 a. What is the meaning of social responsibility of business? Explain Social responsibility of business towards different groups. (10 Marks)
- b. What is a social audit? Explain its benefits and limitations. (10 Marks)

OR

- 6 a. Write a short note on behavioural approach of leadership. (04 Marks)
- b. Explain any four essentials of an effective control system. (08 Marks)
- c. What is corporate governance? List the benefits of Good Corporate Governance. (08 Marks)

Module-4

- 7 a. Briefly explain any four characteristics of a successful entrepreneurship. (10 Marks)
- b. Explain the strategies in capacity building for entrepreneurship. (10 Marks)

OR

- 8 a. Discuss about the business opportunities in India. (10 Marks)
b. Write a short note on Mobility of Entrepreneurs. (10 Marks)

Module-5

- 9 a. Explain the formalities for starting business enterprise. (10 Marks)
b. Discuss the reasons behind the failure of certain business plan. (10 Marks)

OR

- 10 a. Briefly explain about NGO's (Non governmental organizations). (10 Marks)
b. Write a short note on:
(i) TCO's (Technical Consultancy Organizations)
(ii) SIDBI (Small Industries Development Bank of India). (10 Marks)

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Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Microwave Theory and Antennas

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1. a. Derive Transmission line Equations in voltage and current forms. (08 Marks)
 b. List the characteristics of Smith Chart. (07 Marks)
 c. A certain transmission line has a characteristic impedance of $75 + j0.01 \Omega$ and is terminated in a load impedance of $70 + j50 \Omega$, compute
 (i) Reflection co-efficient
 (ii) Transmission co-efficient. (05 Marks)

OR

2. a. With neat diagram, explain the typical Microwave system. (07 Marks)
 b. A transmission line has the following parameters, $R = 2\Omega/m$, $G = 0.5 \text{ mmho/m}$, $f = 1 \text{ GHz}$, $L = 8 \text{ nH/m}$, $C = 0.23 \text{ PF}$.
 Calculate (i) Characteristic impedance (ii) Propagation constant (05 Marks)
 c. With the help of a functional block diagram, explain construction and modes of working of a GUNN Diode. (08 Marks)

Module-2

3. a. Write the S-matrix representation for Mutliport network. (07 Marks)
 b. With a neat diagram, explain the working of precession type variable attenuator. (08 Marks)
 c. A 20 mW signal is fed into one of the collinear Port 1 of a lossless H-plane T-junction. Calculate the power delivered through each port when other ports are terminated in Matched load. (05 Marks)

OR

4. a. State and explain the properties of S-matrix. (07 Marks)
 b. What is Magic-tee? Derive its scattering matrix. (07 Marks)
 c. In an H-plane T-junction, compute power delivered to the loads of 40 ohm and 60 ohm connected to arms-1 and 2 when a 10 mW power is delivered to the matched Port 3. (06 Marks)

Module-3

5. a. Discuss briefly Micro-strip lines and its losses and also derive the expression for Quality factor. (08 Marks)
 b. Define the following terms with respect to antennas:
 (i) Radiation Intensity
 (ii) Beam area.
 (iii) Directivity
 (iv) Beam efficiency. (08 Marks)
 c. A radio link has a 15 W transmitter connected to an antenna of 2.5 m^2 effective aperture at 5 GHz. The receiving antenna has an effective aperture of 0.5 m^2 and is located at 15 km line of sight distance from the transmitting antenna. Assume lossless antennas. Find the power delivered to the receiver. (04 Marks)

OR

- 6 a. A lossless parallel strip line has a conducting strip width W . The substrate dielectric separating the two conducting strips has a relative dielectric constant ϵ_{rd} of 6 and a thickness d of 4 mm. Calculate
- The required width W of the conducting strip in order to have a characteristic impedance of 50Ω .
 - The strip line capacitance.
 - The strip line inductance.
 - The phase velocity of the wave in the parallel strip lines. (08 Marks)
- b. Show that maximum effective aperture of a $\lambda/2$ dipole antenna is $0.13 \lambda^2$. (06 Marks)
- c. Find the Directivity of the following using Exact method,
- $U = U_m \sin^2 \theta \sin^3 \phi$
 - $U = U_m \cos^4 \theta \sin^2 \phi$ where $0 \leq \theta \leq \frac{\pi}{2}$ and $0 \leq \phi \leq 2\pi$. (06 Marks)

Module-4

- 7 a. Derive the expression and draw the field pattern for two isotropic point sources of the same Amplitude and same phase. (08 Marks)
- b. Derive the expression for radiation resistance of short electric dipole. (08 Marks)
- c. Find the length of an Elementary dipole having a radiation resistance of 5Ω at a frequency of 5 MHz. (04 Marks)

OR

- 8 a. Derive array factor expression in case of n isotropic point sources of equal Amplitude and spacing. (07 Marks)
- b. Starting from Electric and Magnetic field potential, obtain the far field components for a short dipole. (10 Marks)
- c. For a short dipole $\frac{\lambda}{15}$ long find the radiation resistance. (03 Marks)

Module-5

- 9 a. Obtain the expression for radiation resistance of small loop antenna. (07 Marks)
- b. Briefly explain helical antenna with its helical geometry. (05 Marks)
- c. Find the length L , H-plane aperture and flare angle θ_E and θ_H of pyramidal horn for which E-plane aperture is 10λ Horn is fed by a rectangular waveguide with TE_{10} mode. Assume $\delta = 0.2\lambda$ in E-plane and 0.375λ in H-plane. Also find beam widths and directivity. (08 Marks)

OR

- 10 a. Derive the expression for strength E_ϕ and H_θ in case of small loop Antenna. (08 Marks)
- b. Explain different types of Horn Antenna. (06 Marks)
- c. Design Yagi-Uda antenna of six elements to provide a gain of 12 dB, if the operating frequency is 200 MHz. (06 Marks)

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CBCS SCHEME

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21EC63

Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 VLSI Design and Testing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Realize the CMOS compound gate for the following functions:
(i) $Y = \overline{ABC + D}$ (ii) $Y = \overline{A(B + C) + DE}$ (08 Marks)
b. Implement a positive edge triggered D flip flop using transmission gate and write the necessary timing diagram. (08 Marks)
c. Analyze the working of tristate inverter. (04 Marks)

OR

- 2 a. Draw the circuit diagram of CMOS inverter and derive its transfer characteristics by graphical method. (06 Marks)
b. Derive the equation for drain current of a MOSFET in non-saturated and saturated region of operation. (10 Marks)
c. Explain the following non-ideal effects of MOSFETs:
(i) Channel length modulation (ii) Mobility degradation. (04 Marks)

Module-2

- 3 a. With neat diagrams, explain the complete CMOS n-well fabrication process. (10 Marks)
b. Draw the layout diagram for the following function and also estimate the area.
 $Y = \overline{(A + B + C)D}$ (10 Marks)

OR

- 4 a. Using Elmore delay model estimate the t_{pdf} and t_{pdr} of a 3-input NAND gate if the output is loaded with 'h' identical gates. (08 Marks)
b. Find the logical effort and parasitic delay of (i) 2 input NOR gate and (ii) 3-input NAND gate. (06 Marks)
c. Construct necessary equivalent circuit for the computation of t_{pdf} of an inverter driving another inverter using RC delay model. (06 Marks)

Module-3

- 5 a. With necessary circuit diagrams explain the operation of (i) 4 transistor DRAM and (ii) 3 transistor DRAM cells. (10 Marks)
b. Explain the operation of full CMOS SRAM cell with necessary circuit topology. (06 Marks)
c. Explain the hysteresis characteristics of ferroelectric capacitor with necessary diagram. (04 Marks)

OR

- 6 a. Explain the operation of 4×4 NOR based ROM array with necessary circuit diagram. (08 Marks)
b. With necessary circuit diagram and bias conditions, explain the operation of NOR flash memory. (06 Marks)
c. Explain binary tree based column decoder design with necessary diagram. (06 Marks)

Module-4

- 7 a. Differentiate between fault and failure with an example. Explain different types of stuck at fault with example. (06 Marks)
- b. Explain feedback bridging fault with an example. (06 Marks)
- c. For the circuit shown in Fig.Q7(c), using Boolean difference (i) detect s-a-0 and s-a-1 at x_1 , (ii) determine partial Boolean difference for $x_2 - 1 - n - p - F$.

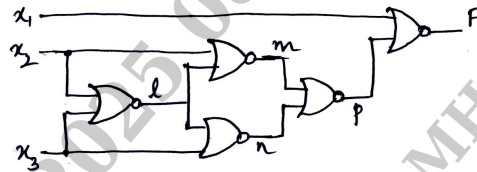


Fig.Q7(c)

(08 Marks)

OR

- 8 a. What is fault diagnosis? Explain delay fault detection with an example. (08 Marks)
- b. Find the test pattern for line 6 s-a-0 for the circuit shown in Fig.Q8(b) using D algorithm.

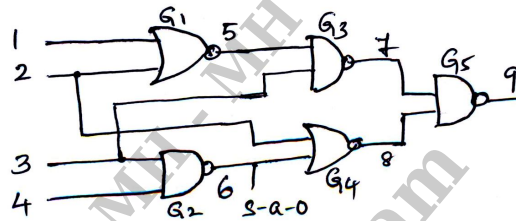


Fig.Q8(b)

(12 Marks)

Module-5

- 9 a. For the state table shown in Table Q9(a), find
(i) Response for 010 sequence, (ii) Homing tree, (iii) Distinguishing tree.

Present State	Input	
	$x = 0$	$x = 1$
A	B, 0	D, 0
B	A, 0	B, 0
C	D, 1	A, 0
D	D, 1	C, 0

Table Q9(a)

(10 Marks)

- b. Write a note on functional fault model to detect faults in sequential circuits. (05 Marks)
- c. Explain the process of testing sequential circuit as iterative combinational circuits. (05 Marks)

OR

- 10 a. Define the terms controllability and observability with an example. (05 Marks)
- b. With a neat logic diagram, explain clocked hazard free latches used in LSSD technique. (08 Marks)
- c. Explain partial scan technique using system clock with necessary diagram. (07 Marks)

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CBCS SCHEME

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21EC641

Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Artificial Neural Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. List some of the Application scope of Neural network? (06 Marks)
- b. With a neat schematic diagram explain Biological neuron. (06 Marks)
- c. Compare Biological neural network and Artificial neural network. (08 Marks)

OR

- 2 a. Discuss in detail the McCulloch – Pitts neuron model. (05 Marks)
- b. Define ANN. Explain Artificial neuron model architecture, provide the equation. (05 Marks)
- c. What is meant by Activation function? Explain different types of Activation function used in neural network. (10 Marks)

Module-2

- 3 a. Write and explain the flowchart and algorithm for Hebb training network. (10 Marks)
- b. Design a Hebb network to implement logical AND function (use bipolar input and targets). (10 Marks)

OR

- 4 a. Define Adaptive linear neuron? Explain architecture, flow chart for training process. (10 Marks)
- b. Discuss the perceptron architecture and algorithm for training network for single output classes. (10 Marks)

Module-3

- 5 a. Explain the concept of Back propagation network with the help of Architecture and training algorithm. (10 Marks)
- b. Write the architecture and algorithm steps for radial basis function network training process. (10 Marks)

OR

- 6 a. What are the different learning factors of BPN? (06 Marks)
- b. Write the testing algorithm for BPN. (06 Marks)
- c. Explain the concept of Time Delay neural networks. (08 Marks)

Module-4

- 7 a. Describe the training algorithm and flowchart for pattern association memory network using Hebb rule. (10 Marks)
- b. Define Auto Associative memory network? Explain Architecture and flowchart for training process. (10 Marks)

OR

- 8 a. Write a note on outer product rule. (04 Marks)
b. Write and explain the testing and training algorithm for discrete Hopfield network. (08 Marks)
c. Discuss the architecture and testing algorithm for hetero-associative memory network. (08 Marks)

Module-5

- 9 a. Explain Kohonen self organizing feature maps using linear array of cluster units. (10 Marks)
b. Describe the algorithm steps for Mexican Hat net. (10 Marks)

OR

- 10 a. Write Maxnet architecture and explain testing / applications algorithm of Maxnet. (08 Marks)
b. Define features mapping? Explain in detail 1D and 2D feature mapping network. (08 Marks)
c. Explain the concept of vector quantization with a neat architecture diagram. (04 Marks)

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CBCS SCHEME

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21EC643

Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Python Programming

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain for loop with respect to the range function with the help of an example. (08 Marks)
- b. List out the boolean and comparison operators with their functions. (07 Marks)
- c. Write a program to accept only the integer from the user and to classify the accepted input as even or odd number. Include try and except for input validation. (05 Marks)

OR

- 2 a. List any three built-in functions. Explain the definition and function call of the user defined functions along with parameter and return statements with example. (08 Marks)
- b. List out the rules to be followed while creating the variables. Explain the significance of global statement with an example. (07 Marks)
- c. Write a program to find the factorial of a number using function. (05 Marks)

Module-2

- 3 a. Explain the following with respect to lists with an example for each:
(i) Negative indexing (ii) Slicing
(iii) Concatenation (iv) Replication (08 Marks)
- b. Explain the following dictionary methods with an example:
(i) keys(), values(), items()
(ii) get()
(iii) setdefault() (07 Marks)
- c. Write a program using lists to store N numbers accepted from the user and to display the list average. (05 Marks)

OR

- 4 a. Explain any four list methods with an example for each. (08 Marks)
- b. Explain all the string methods with an example for each. (07 Marks)
- c. Write a program to create a dictionary of 10 items (USN: Marks) pairs by accepting the inputs from the user. Display the item with the maximum and minimum marks. (05 Marks)

Module-3

- 5 a. List out the steps involved in matching a regular expression. (08 Marks)
- b. With the help of an example explain the variable saving process with shelve module. (07 Marks)
- c. Write a program to find the total size of all the files in a directory. (05 Marks)

OR

- 6 a. Briefly explain character classes in Regular expression module. (08 Marks)
- b. Explain the methods related to the absolute and relative paths. (07 Marks)
- c. Write a program to search for lines having '@' sign between characters in a read text file. (05 Marks)

Module-4

- 7 a. Explain pure and modifier functions with an example for each. (08 Marks)
b. Explain operator overloading with an example. (07 Marks)
c. Write a program that uses class to store the name and marks of students. Use list to store the marks in 3 subjects. (05 Marks)

OR

- 8 a. Explain classes, objects and attributes with an example. Show that objects are mutable with the help of a snippet. (08 Marks)
b. Explain `__init__` and `__str__` methods with an example. (07 Marks)
c. Write a program to create two objects of the class point with x and y coordinates as attributes. Pass these two objects to a function which computes and displays the distance between the two points. (05 Marks)

Module-5

- 9 a. Explain HTML parsing using BeautifulSoup with the help of an example. (08 Marks)
b. Explain XML parsing with the help of an example. (07 Marks)
c. Write a program to retrieve a text document from the web using urllib. (05 Marks)

OR

- 10 a. Explain JSON parsing with the help of an example. (08 Marks)
b. Explain the creation of a database table with the help of an example. (07 Marks)
c. Briefly explain the three kinds of keys used in database model. (05 Marks)

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Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Advanced Linear Algebra

Time: 3 hrs.

Max. Marks: 100

Note : Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the vector space with examples. (06 Marks)
- b. Test the consistency and solve them $x - 3y + 4z = -4$, $3x + 7y + 7z = -8$ and $-4x + 6y - z = 7$. (07 Marks)
- c. Solve the linear system of equations $3x + 2y + 7z = 4$, $2x + 3y + z = 5$, $3x + 4y + z = 7$ using LU - decomposition. (07 Marks)

OR

- 2 a. Define the terms : i) Subspace ii) Bases and iii) Dimensions. (06 Marks)
- b. Prove that intersection of two subspaces of a V.S $V(F)$ is a subspace of $V(F)$. (07 Marks)
- c. Show that the set $S = \{V_1, V_2, V_3, V_4\}$ where $V_1 = (1, 0, 1, 0)$, $V_2 = (0, 1, -1, 2)$, $V_3 = (0, 2, 2, 1)$, $V_4 = (1, 0, 0, 1)$. (07 Marks)

Module-2

- 3 a. Define the terms : i) Linear transformation ii) Rank iii) Nullity. (06 Marks)
- b. Let v and w be V.S $V(F)$ and let T be a L.T from v into w and let v be a finite dimension. Prove that $\text{rank}(T) + \text{nullity}(T) = \dim v$. (07 Marks)
- c. Show that the mapping $T : R^3 \rightarrow R^3$ defined by $T(x, y, z) = (x + 1, y, z)$ are linear transformation. (07 Marks)

OR

- 4 a. Let v and w be V.S. $V(F)$ and let $T : v \rightarrow w$. Prove that T is invertible and T^{-1} is also invertible. (06 Marks)
- b. Show that the vectors $V_1 = (1, 1, 0, 0)$, $V_2 = (2, 0, 1, 0)$, $V_3 = (0, 1, 2, -1)$, $V_4 = (0, 1, -1, 0)$ forms a basis for R^4 . Find the coordinates of each of the standard basis vectors in the ordered basis $\{V_1, V_2, V_3, V_4\}$.. (07 Marks)
- c. Let $T : V \rightarrow W$ is a Linear transformations from the $V(F)$ to $W(F)$. Prove that
i) $T(0) = 0$, ii) $T(-x) = -T(x)$ iii) $T(x - y) = T(x) - T(y)$. (07 Marks)

Module-3

- 5 a. Define i) Inner product space ii) Orthogonal sets iii) Orthogonal projections. (06 Marks)
- b. Let $u_1 = \begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix}$, $u_2 = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$ and $y = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ prove that y as the sum of vector in w and a vector orthogonal to w . (07 Marks)

- c. Show that $\{V_1, V_2, V_3\}$ is an orthonormal basis of $\mathbb{R}^3$, where

$$V_1 = \begin{bmatrix} \frac{3}{\sqrt{11}} \\ \frac{1}{\sqrt{11}} \\ \frac{1}{\sqrt{11}} \end{bmatrix}, \quad V_2 = \begin{bmatrix} \frac{-1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}, \quad V_3 = \begin{bmatrix} \frac{-1}{\sqrt{66}} \\ \frac{-4}{\sqrt{66}} \\ \frac{7}{\sqrt{66}} \end{bmatrix} \quad (07 \text{ Marks})$$

OR

- 6 a. Use Gram – Schmidt Orthogonalization process to vectors $(1, 0, 1)$, $(1, 0, -1)$ and $(0, 3, 4)$ to obtain an orthonormal basis $\{\alpha_1, \alpha_2, \alpha_3\}$ for $\mathbb{R}^3$. (10 Marks)
- b. Obtain the QR – decomposition for

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}. \quad (10 \text{ Marks})$$

Module-4

- 7 a. Find the Eigen values and Eigen vectors of $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$. (06 Marks)

- b. Obtain the Singular value decomposition of $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 3 \end{bmatrix}$. (07 Marks)

- c. Diagonalize the following matrix, if possible $A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$. (07 Marks)

OR

- 8 a. Explain the quadratic forms. (06 Marks)
- b. Explain the constrained optimization. (07 Marks)
- c. Explain the singular value decomposition. (07 Marks)

Module-5

- 9 a. Define the graphs and networks. (06 Marks)

- b. Draw the multigraph G whose adjacency matrix $M_A = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}$. (07 Marks)

- c. Draw the undirected graphs K_3 and K_5 . (07 Marks)

OR

- 10 a. What are the applications of matrices in Engineering? (10 Marks)
- b. Describe the Computer Graphics. (10 Marks)

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Fourth Semester B.E. Degree Examination, Dec.2024/Jan.2025

Analog Circuits

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Derive the expression for Emitter current of a voltage divider bias and also discuss how to make I_E insensitive to variation in β and temperature. (10 Marks)
- b. Design a collector to Base feedback resistor bias to obtain a dc current of 1 mA and to ensure $\pm 2V$ signal swing at the collector with $V_{CE} = 2.3V$. Assume $V_{CC} = 10V$ and $\beta = 100$. (07 Marks)
- c. What is trans-conductance of BJT and mention its significance? (03 Marks)

OR

- 2 a. Obtain the following expression of a BJT of small signal analysis.
 - i) Total instantaneous collector current
 - ii) Input resistance at the base(10 Marks)
- b. Discuss the following biasing scheme used in MOS
 - i) By fixing V_{GS}
 - ii) By fixing V_{GS} and connecting a resistance in the source.(10 Marks)

Module-2

- 3 a. Discuss the basic configuration of MOSFET. (06 Marks)
- b. For a common source amplifier shown in Fig Q3(b), determine R_{in} , AV_0 , R_0 and G_v

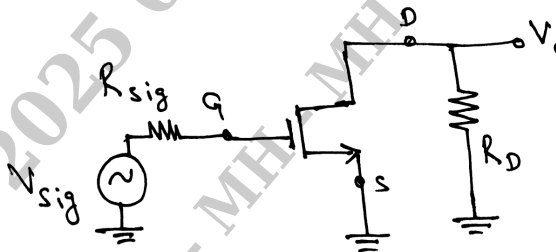


Fig Q3(b)

(14 Marks)

OR

- 4 a. For an n-channel MOSFET with $t_{ox} = 10nm$, $L = 1.0 \mu m$, $W = 10 \mu m$, $L_{OV} = 0.05 \mu m$, $C_{sbo} = C_{dbo} = 10 fF$, $V_0 = 0.6V$, $V_{SB} = 1V$ and $V_{DS} = 2V$. Calculate C_{ox} , C_{ov} , C_{gs} , C_{gd} , C_{sb} and C_{db} . (10 Marks)
- b. Explain the working of FET based phase shift oscillator and also mention the necessary conditions for sustained oscillation. (10 Marks)

Module-3

- 5 a. Explain the following properties of Negative. Feedback.
 i) Gain, De-sensitivity ii) Bandwidth Extension iii) Noise reduction (14 Marks)
- b. A negative feedback amplifier has a $A_f = 100$ and $A = 10^5$. What is the feedback factor? If a manufacturing error results in a reduction of A to 10^3 , what is the closed loop voltage Gain? What is the percentage change in A_f ? (06 Marks)

OR

- 6 a. Explain the working of class B output stage. (08 Marks)
- b. For emitter follower Class A output stage $V_{cc} = 10V$, $I = 100$ mA and $R_L = 100 \Omega$. If the output voltage is an 8 V – peak sinusoid, find :
 i) Power delivered to load
 ii) Average power drawn from the supplies
 iii) Power conversion efficiency ignore the loss on Q_3 and R . (06 Marks)
- c. Explain how cross over distortion can be eliminated to class AB output stage. (06 Marks)

Module-4

- 7 a. For the voltage Seri feedback amplifier, derive the expressions of
 i) Exact voltage Gain ii) Input resistance with feedback iii) Output resistance with feedback (14 Marks)
- b. For the inverting amplifier $R_1 = 470 \Omega$ and $R_F = 4.7 K\Omega$. Assume $A = 200000$, $R_i = 2 M\Omega$, $R_o = 75 \Omega$ and $f_o = 5$ Hz. Calculate A_F , R_{iF} , R_{oF} and f_F . (06 Marks)

OR

- 8 a. Explain the working of instrumentation amplifier using Transducer bridge with necessary equations. (08 Marks)
- b. Explain the working of Inverting Schmitt trigger with input and output waveforms. (08 Marks)
- c. For a Differential configuration summer $R = 1 K\Omega$, $V_a = 2V$, $V_b = 3V$, $V_c = 4V$, $V_d = 5V$ and supply voltage of $\pm 15V$. Determine the output voltage V_o . (04 Marks)

Module-5

- 9 a. Derive the expression of output voltage of a 4-bit Binary weighted resistor type DAC. Mention its disadvantages. (10 Marks)
- b. Draw the block diagram of successive approximation ADC and explain it. (10 Marks)

OR

- 10 a. Explain the working of First order active Lowpass filter with the help of magnitude voltage gain and also design to get a cutoff frequency of 1 KHz with a passband gain of 2. (10 Marks)
- b. Explain the working of Astable multi-vibration using 555 Timer and also derive the expression of Frequency of oscillation. (10 Marks)

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Fourth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Control Systems

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Use of semilog graph sheets are permitted.

Module-1

- 1 a. Define control system. Distinguish between open loop and closed loop systems with examples. (07 Marks)
- b. For the mechanical system shown in Fig. Q1 (b),
 - (i) Draw mechanical network.
 - (ii) Write difference equations of performance.
 - (iii) Draw electrical network based on force voltage analogy. (08 Marks)

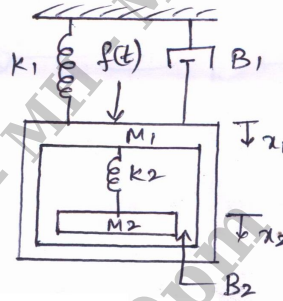


Fig. Q1 (b)

- c. Discuss the effect of feedback on,
 - (i) Overall gain
 - (ii) Stability (05 Marks)

OR

- 2 a. For the electromechanical system shown in Fig. Q2 (a), determine the transfer function $\frac{X(s)}{E(s)}$. (10 Marks)

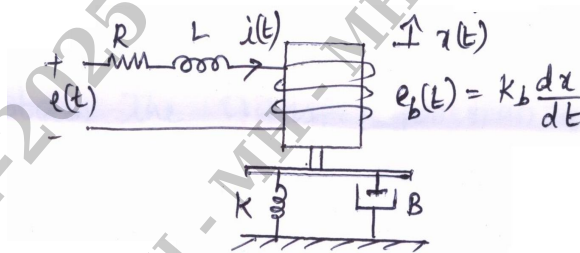


Fig. Q2 (a)

- b. For the mechanical system shown in Fig. Q2 (b),
 - (i) Draw the mechanical network
 - (ii) Draw electrical network based on torque-current analogy
 - (iii) Write performance equations. (10 Marks)

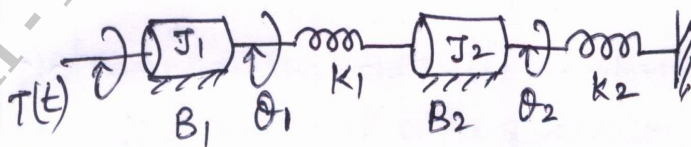


Fig. Q2 (b)

Module-2

- 3 a. Write Mason's gain formula for signal flow graph. Indicate what each term represents. (05 Marks)
- b. Reduce the block diagram shown in Fig. Q3 (b), using block diagram reduction rules and obtain $\frac{C(s)}{R(s)}$. (08 Marks)

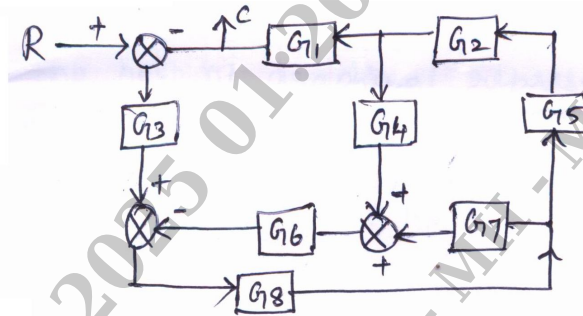


Fig. Q3 (b)

- c. Find the transfer function for the following network shown in Fig. Q3 (c) using Mason's gain formula.

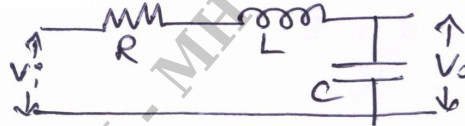


Fig. Q3 (c)

(07 Marks)

OR

- 4 a. Obtain the transfer function of the system shown in Fig. Q4 (a) using Mason's gain formula. (12 Marks)

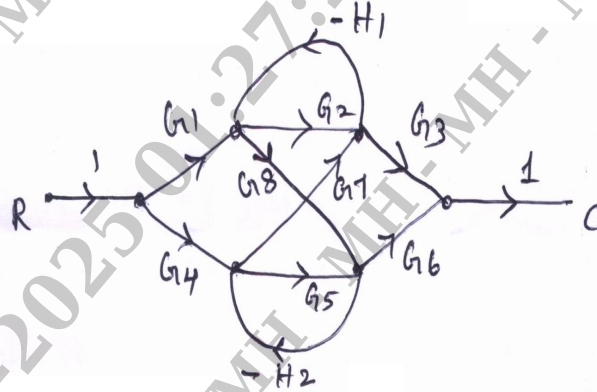


Fig. Q4 (a)

- b. Determine the overall transfer function for the block diagram shown in Fig. Q4 (b).

(08 Marks)

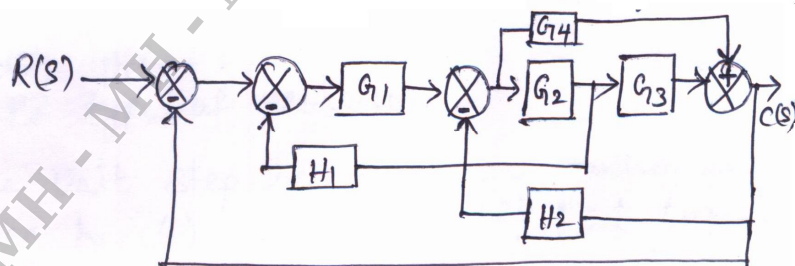


Fig. Q4 (b)

Module-3

- 5 a. With the help of graphical representation and mathematical expression, explain the following test signals : (i) Step signal (ii) Ramp signal (iii) Parabolic signal (iv) Impulse signal. (08 Marks)
- b. For a unity negative feedback control system with $G(s) = \frac{50}{s(s+5)}$, find the following :
 (i) Percentage overshoot for unit step input.
 (ii) Settling time for a unit step input.
 (iii) Steady state error for an input defined by polynomial $r(t) = 2 + 4t + 6t^2$; $t \geq 0$. (08 Marks)
- c. Define rise time and maximum overshoot and also write their formula for II order systems. (04 Marks)

OR

- 6 a. For a unity feedback control systems, the open loop transfer function $G(s) = \frac{10(s+2)}{s^2(s+1)}$ find,
 (i) The position, velocity and acceleration error constants.
 (ii) The steady state error when input is $R(s)$ where $R(s) = \frac{3}{s} - \frac{2}{s^2} + \frac{1}{3s^3}$ (08 Marks)
- b. With the help of general block diagrams, explain the following :
 (i) PD type of controller.
 (ii) PI type of controller. (08 Marks)
- c. The unit step response of a system is given by $C(t) = \frac{5}{2} + 5t - \frac{5}{2}e^{-2t}$. Find transfer function and identify order of system. (04 Marks)

Module-4

- 7 a. The open loop transfer function of unity feedback system is given by,
 $G(s) = \frac{K(s+1)}{s^3 + as^2 + 2s + 1}$. Determine the value of K and a so that system oscillates at frequency of 2 rad/sec. (08 Marks)
- b. State and explain Routh's stability criterion for determining the stability of the system and mention its limitations. (06 Marks)
- c. Sketch the root locus plot for a negative feedback control system having an open loop transfer function,
 $G(s)H(s) = \frac{K}{s(s+1)(s+2)}$. (06 Marks)

OR

- 8 a. The open loop transfer function of a system is $G(s) = \frac{K}{s(1+s)(1+0.1s)}$. Determine the values of K such that,
 (i) Gain margin = 10 dB
 (ii) Phase margin = 24° .
 Use Bode plot. (12 Marks)
- b. Define the following terms in connection with bode plots :
 (i) Gain cross over frequency
 (ii) Phase cross over frequency
 (iii) Gain margin
 (iv) Phase margin (08 Marks)

Module-5

- 9 a. Sketch the polar plot for open loop transfer function, $G(s)H(s) = \frac{10}{(s+2)(s+4)}$. Determine gain cross over frequency, phase cross over frequency, gain margin, phase margin. Also comment on stability. (10 Marks)
- b. Explain Nyquist stability criterion and also list the advantages of Nyquist plot. (05 Marks)
- c. Write a short note on lead compensator. (05 Marks)

OR

- 10 a. List the properties of state transition matrix. (05 Marks)
- b. Obtain state transition matrix $\phi(t)$ of the following system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$
 Also obtain the inverse of state transition matrix $\phi^{-1}(t)$. (10 Marks)
- c. Define : (i) State variables
 (ii) State vector
 (iii) State space (05 Marks)

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Fourth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Engineering Statistics and Linear Algebra

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define CDF of a Random variable. Mention its properties and types. (10 Marks)
- b. Given the data in the Table Q1(b)
- Plot the PDF and CDF of the discrete random variable
 - Write expressions for $f_X(x)$ and $F_X(x)$ using unit delta functions and unit step functions :

x	x_a	x_b	x_c	Total
$P[X = x]$	0.24	0.32	0.44	1

Table Q1(b)

(10 Marks)

OR

- 2 a. Summarize the properties of PDF. Prove that the total area under PDF curve is unity. (10 Marks)
- b. Given the data in the Table Q2(b).
- What are the mean and variance of 'X'
 - If $Y = X^2 + 2$, what are μ_y and σ_y^2 .

K	1	2	3	4	5
X_k	2.1	3.2	4.8	5.4	6.9
$P(x_k)$	0.21	0.18	0.2	0.22	0.19

Table Q2(b)

(10 Marks)

Module-2

- 3 a. Explain the following with respect to Bivariate Random variable.
- Correlation
 - Covariance
 - Uncorrelated X and Y
 - Orthogonal X and Y
 - Independent X and Y.
- (10 Marks)
- b. Let X is a random variable, $\mu_x = 4$ and $\sigma_x = 5$ and Y is a random variable, $\mu_y = 6$ and $\sigma_y = 7$. The correlation coefficient is -0.7. If $U = 3x + 2y$, what are
- Var [U]
 - CoV [UX]
 - CoV [UY].
- (10 Marks)

OR

- 4 a. Briefly explain the following random variables
- Chi-square RV
 - Student-T RV
 - Cauchy RV
 - Rayleigh RV.
- (10 Marks)
- b. The joint PDF $f_{XY}(x, y) = C$, a constant when $(0 < x < 3)$ and $(0 < y < 3)$ and is '0' otherwise.
- What is the value of constant C
 - What is the PDF's for X and Y
 - What is $F_{XY}(x, y)$ when $(0 < x < 3)$ and $(0 < y < 3)$
 - What are $F_{XY}(x, \infty)$ and $F_{XY}(\infty, y)$
 - Are X and Y independent?
- (10 Marks)

Module-3

- 5 a. Interpret the following with respect to random process
- Random process
 - Ensemble
 - PDF
 - Independence
 - Expectations
 - Stationary.
- (12 Marks)

- b. The magnitude of a zero mean white noise spectrum is $K = 3.6 \times 10^{-8} \text{ V}^2\text{-S}$. This noise is the input to a low pass RC circuit: $R = 38 \text{ k}\Omega$, $C = 0.1 \text{ }\mu\text{F}$. Find the networks output PSD, $S_y(w)$. (08 Marks)

OR

- 6 a. Discuss the Auto correlation and cross correlation functions with their properties. (12 Marks)
- b. A Random process is described by $y(t) = A \cos(w_c t + \theta)$ where A and w_c are constants, but θ is a random variable distributed uniformly between $\pm\pi$. Determine :
- PDF of random variable ' θ '
 - Mean of $y(t)$
 - Auto correlation function $R_y(\tau)$
 - Mean power and Auto variance of $y(t)$

(08 Marks)

Module-4

- 7 a. Illustrate vector space with its properties in detail. (08 Marks)
- b. Apply Gram-Schmidt process to

$$a = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad b = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad c = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Write the result in the form of $A = QR$.

(12 Marks)

OR

- 8 a. Outline the four fundamental subspaces of matrices. (08 Marks)
- b. Determine : i) matrix U and Rank ii) $\text{rref}(R)$ iii) Null space of matrix and identify free variables in null space for the matrix given $A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}$ (12 Marks)

Module-5

- 9 a. Define determinants with its properties in detail. (13 Marks)

- b. Determine the Eigen values of matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ (07 Marks)

OR

- 10 a. Diagonalize the matrix $A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ and hence find A^4 , Also find the matrix 'P' such that $P^{-1}AP$ is diagonal. (14 Marks)

- b. Reduce the matrix A to U and find $\det A$ using pivots of A .

$$A = \begin{bmatrix} 2 & 5 & 3 \\ 1 & 2 & 4 \\ -1 & 3 & 6 \end{bmatrix}$$

(06 Marks)

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CBCS SCHEME

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18EC45

Fourth Semester B.E. Degree Examination, Dec.2024/Jan.2025

Signals and Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Check whether the following signals are periodic or not. If periodic find the fundamental period.

i) $x(t) = \sin\left(\frac{\pi}{3}t\right) + 2 \cos\left(\frac{8\pi}{3}t\right)$

ii) $x[n] = \sin\left(\frac{3\pi n}{4}\right) \sin\left(\frac{\pi}{3}n\right)$

iii) $x(t) = e^{j\frac{12\pi}{7}t} + e^{j\frac{12\pi}{5}t}$

(10 Marks)

- b. Calculate the energy and power of the following signals as applicable:

i) $x[n] = (j)^n + (j)^{-n}$

ii) $x[n] = 8(0.5)^n U[n]$

(06 Marks)

- c. Determine and sketch the even and odd parts of the signal depicted in Fig.Q.1(c). (04 Marks)

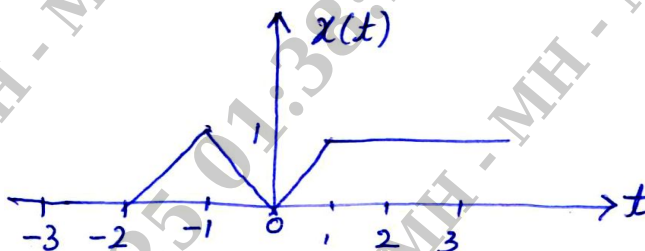


Fig.Q.1(c)

OR

- 2 a. A trapezoidal pulse $x(t)$ is defined by

$$x(t) = \begin{cases} 5-t, & 4 \leq t \leq 5 \\ 1, & -4 \leq t \leq 4 \\ t+5, & -5 \leq t \leq -4 \\ 0, & \text{otherwise} \end{cases}$$

is applied to a differentiator having the input-output relation $y(t) = \frac{dx(t)}{dt}$. Find the energy of $y(t)$. (04 Marks)

- b. Show that the product of two even signals or two odd signals is an even signal, while the product of an even and odd signal is an odd signal. (06 Marks)

c. From the signals indicated in Fig.Q.2(c), derive the following signals:

- $x(t-1) y(-t)$
- $x(t) [\delta(t-1) + \delta(t-2)] + y(t) s(t+2)$
- $x(t) y(t-1)$
- $x(t) y(-t-1)$

(10 Marks)

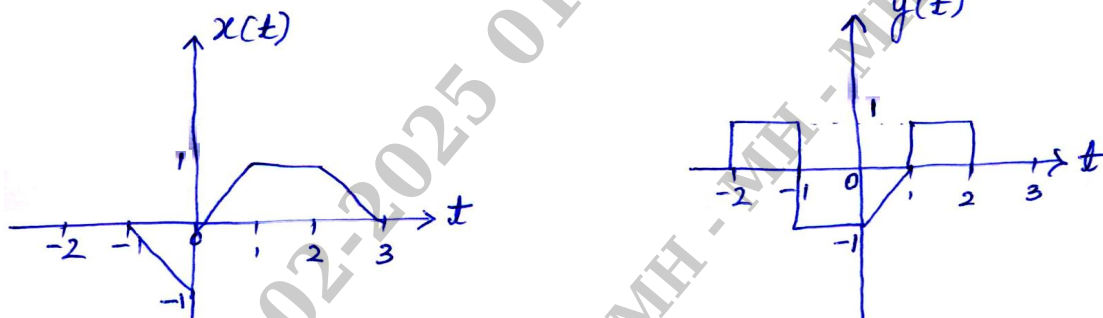


Fig.Q.2(c)

Module-2

- 3 a. Determine whether the system described by the following input-output relation is
i) Linear ii) Causal iii) Time-invariant iv) Memoryless v) Stable.

$$y(t) = 2x(t) + 3$$

(06 Marks)

- b. Perform the convolution operation on the following signals:

$$x(t) = e^{-2t} u(t)$$

$$h(t) = \mu(t+2)$$

(08 Marks)

- c. Show that

$$i) [x(n) * h_1(n)] * h_2(n) = x(n) * [h_1(n) * h_2(n)]$$

$$ii) x(n) * h(n) = h(n) * x(n)$$

(06 Marks)

OR

- 4 a. Given $x[n] = \alpha^n U[n]$ and $h[n] = \beta^n U[n]$, perform $x[n] * h[n]$. (08 Marks)

- b. Determine whether the following systems represented by input-output relations are invertible. If invertible then represent their inverse system

$$i) y(t) = s^{t_0} \{x(t)\} \quad ii) y(t) = \frac{1}{L} \int_{-\infty}^t x(z) dz$$

$$iii) y(t) = x^2(t) \quad iv) y(t) = 2x(t)$$

(08 Marks)

- c. Perform convolution operations on the following signals and sketch the output:

$$x[n] = \delta(n+1) + 2\delta(n) + 3\delta(n-1) - 2\delta(n-2) + \delta(n-3)$$

$$h[n] = \left(\frac{1}{2}\right)^n [U[n] - U[n-4]]$$

(04 Marks)

Module-3

- 5 a. Determine whether the following systems represented by impulse responses are stable, causal and memoryless.

$$i) h(n) = U(n-1) - U(n-5)$$

$$ii) h(t) = e^{-t} \mu(+t)$$

$$iii) h(n) = 0.5^{|n|}$$

$$iv) h(t) = \mu(t-1)$$

(08 Marks)

- b. Show that step response of an LTI system is running integral of impulse response. (04 Marks)
- c. Evaluate the Fourier series representation for the signal $x(t) = \sin(2\pi t) + \cos(3\pi t)$. Sketch the magnitude and phase spectra. (08 Marks)

OR

- 6 a. State and prove convolution property for continuous time Fourier series. (04 Marks)
- b. Find the Fourier series representation for the given signal and draw its magnitude and phase spectra. (10 Marks)

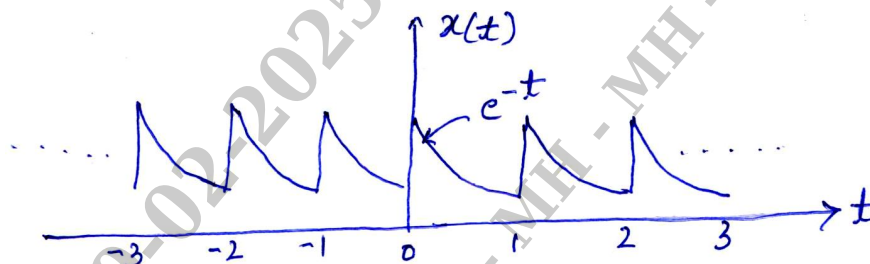


Fig.Q.6(b)

- c. Find the unit step response of the systems given by their impulse responses

i) $h(n) = \left(\frac{1}{2}\right)^n U(n)$

ii) $h(t) = e^{-|t|}$

(06 Marks)

Module-4

- 7 a. State and prove the following properties applicable to continuous time Fourier transform
i) Time shift ii) Frequency differentiation (06 Marks)
- b. Find the continuous time Fourier transform of $x(t) = e^{-at} u(t)$; $a > 0$. Draw its magnitude and phase spectra. (08 Marks)
- c. Find the discrete time Fourier transform of the following signals:
i) $x(n) = (-1)^n U(n)$
ii) $x(n) = a^{|n|}$. (06 Marks)

OR

- 8 a. State and prove Parseval's theorem with respect to discrete time Fourier transform and indicate the importance of it. (06 Marks)
- b. The discrete time Fourier transform of a real signal $x(n)$ is $X(\Omega)$. How is the discrete time Fourier transform of the following signals related to $X(\Omega)$.
i) $y_1(n) = x(-n)$
ii) $y_3(n) = x(n) * x(-n)$
iii) $y_2(n) = (1 + \cos n\pi) x(n)$
iv) $y_4(n) = (-1)^{n/2} x(n)$. (08 Marks)
- c. Find the continuous time Fourier transform of
i) $x(t) = \cos(\omega_0 t)$
ii) $x(t) = \begin{cases} 1; & -T < t < T \\ 0; & \text{otherwise} \end{cases}$ (06 Marks)

Module-5

- 9 a. Find the Z-transform of the following sequences and plot its ROC.

i) $x(n) = \left(\frac{1}{2}\right)^n U(n-2)$

ii) $x(n) = 3\left(\frac{-1}{2}\right)^n U(n) - 2[3^n U(-n-1)]$

(06 Marks)

- b. If $x(n)$ is causal, then prove that

i) $x(0) = \lim_{Z \rightarrow \infty} Z X(z)$

ii) $x(\infty) = \lim_{Z \rightarrow 1} [X(Z)(Z-1)]$

(08 Marks)

- c. A LTI system is given by the system function

$$H(Z) = \frac{3 - 4Z^{-1}}{(1 - 3.5Z^{-1} + 1.5Z^{-2})}$$

Specify the ROC of $H(z)$ and $h(n)$ for the following conditions:

- i) The system is stable
ii) The system is causal.

(06 Marks)

OR

- 10 a. Use the properties of Z-transform to find the Z-transform of the following:

i) $a^{-n}U(-n)$

ii) $\left(\frac{1}{2}\right)^n U(n) * \left(\frac{1}{3}\right)^n U(n)$

iii) $\left(\frac{1}{3}\right)^n U(n) + \left(-\frac{1}{2}\right)^n U(n)$

(08 Marks)

- b. Find the inverse Z-transform of

$$X(Z) = \frac{Z(Z^2 - 4Z + 5)}{(Z-3)(Z-2)(Z-1)} \text{ for the ROC } 2 < |Z| < 3 \text{ using partial fraction method. (06 Marks)}$$

- c. Determine the impulse response $h(n)$ and system function $H(z)$ of the system that gives the output

$$y(n) = \left(\frac{1}{3}\right)^n U(n) \text{ for an input } x(n) = \left(\frac{1}{2}\right)^n U(n) - \frac{1}{4}\left(\frac{1}{2}\right)^{n-1} U(n-1).$$

(06 Marks)

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18EC61

Sixth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Digital Communication

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Hilbert transform. What are the applications of Hilbert transform? Prove that a signal $g(t)$ and its Hilbert transform $\hat{g}(t)$ are orthogonal over the entire time interval $(-\infty, \infty)$. (10 Marks)
- b. Derive the expression for the complex low pass representation of band pass system. (10 Marks)

OR

- 2 a. Express bandpass signal $s(t)$ in canonical form. Also explain the scheme for deriving the inphase and quadrature components of the bandpass signal $s(t)$. (10 Marks)
- b. For a binary sequence 0100000001011 construct:
(i) RZ bipolar format (ii) Manchester format (iii) B3ZS format
(iv) B6ZS format (v) HDB3 format (10 Marks)

Module-2

- 3 a. Explain the geometric representation of set of M energy signals as linear combination of N orthonormal basis functions. Illustrate for the case $N = 2$ and $M = 3$ with necessary diagrams and expressions. (10 Marks)
- b. Obtain the maximum likelihood decision rule for the signal detection problem. (10 Marks)

OR

- 4 a. Apply Gram-Schmidt procedure to obtain an orthonormal basis for the signals $s_1(t)$, $s_2(t)$ and $s_3(t)$ as shown in Fig.Q4(a). Also express each of these signals in terms of the set of basis functions.

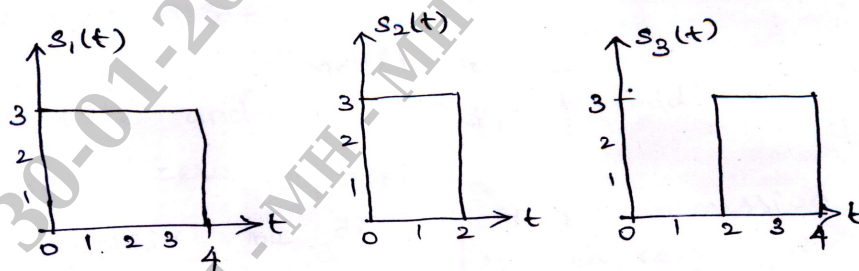


Fig.Q4(a)

- b. Explain the correlation receiver and matched filter receiver with relevant diagrams.

(10 Marks)
(10 Marks)

Module-3

- 5 a. Derive the expression for average probability of error for FSK using coherent detection. Explain transmitter and coherent receiver of FSK. (10 Marks)
- b. With a neat block diagram, explain the generation and optimum detection of DPSK signals. (10 Marks)

OR

- 6 a. Using block diagram, explain the generation and detection of QPSK signal. (10 Marks)
 b. Derive the expression for error probability of binary PSK using coherent detection. (06 Marks)
 c. What is the advantage of M-ary QAM over M-ary PSK system? Obtain the constellation of QAM for $M = 4$ and draw signal space diagram. (04 Marks)

Module-4

- 7 a. State and prove Nyquist condition for zero ISI. (08 Marks)
 b. Explain the digital PAM transmission system. Also derive the expression for Inter Symbol Interference (ISI). (08 Marks)
 c. With neat diagram and relevant expression, explain the concept of adaptive equalization. (04 Marks)

OR

- 8 a. What is a zero forcing equalizer? With a neat block diagram, explain the operation of linear transversal filter. (08 Marks)
 b. Explain the need for precoder in a duobinary signaling. The binary data 001101001 are applied to the input of a duobinary system. Construct the duo binary coder output and corresponding receiver output. Assume that precoder is used. (08 Marks)
 c. Write a note on eye diagram. (04 Marks)

Module-5

- 9 a. Explain the model of a spread spectrum digital communication system. (06 Marks)
 b. Illustrate the working of direct sequence spread spectrum transmitter and receiver with block diagram, waveforms and expressions. (10 Marks)
 c. What is a PN sequence? What are the properties of maximum length sequences? (04 Marks)

OR

- 10 a. Explain frequency hop spread spectrum with neat block diagram. (10 Marks)
 b. Illustrate the CDMA system forward link based on IS-95. (10 Marks)

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18EC62

Sixth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Embedded Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Describe the various units designed in the ARM Cortex M3 architecture, with a neat diagram. (10 Marks)
- b. Explain the exceptions and IRQ interrupts of ARM Cortex M3 with a table and write the 2 timing diagrams when exception and interrupts are to be handled in privileged modes only, during the execution of a privileged or user thread. (10 Marks)

OR

- 2 a. Describe the general purpose Registers, $R_{13} - R_{15}$, special registers and PSR in detail, with the relevant diagrams for all. (10 Marks)
- b. Explain the reset sequence and PUSH and POP instructions with examples, along with relevant diagrams for both. (10 Marks)

Module-2

- 3 a. Describe IF-THEN-ELSE, shift and rotate instructions (5 types) and logic operation instruction (4 types) with examples. (11 Marks)
- b. Explain the following instruction with an example for each MLA, REV, STR, MSR. (04 Marks)
- c. Develop an assembly level language program to add the decimal integer numbers 1 to 20. The result should be stored in register R_9 , with counter as R_8 . (05 Marks)

OR

- 4 a. Describe the ARM ALP/HLP development tools and the organization of CMSIS with diagrams. (11 Marks)
- b. Explain the following instructions with an example for each: MRS, LDR, ADC, NEG. (04 Marks)
- c. Develop a 'C' language program to switch ON or OFF a RED lamp with a delay program having a maximum count of 100 in decimal, for ARM cortex M3 processor. ON and OFF time should be equal. (05 Marks)

Module-3

- 5 a. Explain the four classifications of embedded system with features and examples based on the generations. (04 Marks)
- b. Describe the elements of an embedded system with a block diagram. Also, explain any four types of ROMs. (12 Marks)
- c. Differentiate between RISC and CISC architectures, with respect to their features in the design. (04 Marks)

OR

- 6 a. Explain any four purposes of embedded systems, with suitable examples for each. (04 Marks)
b. Describe optocoupler, DRAM and SPI with diagrams. (12 Marks)
c. Differentiate between embedded and general computing systems with respect to the design based on the applications. (04 Marks)

Module-4

- 7 a. Explain any four characteristics of embedded systems. (04 Marks)
b. Develop and explain the sequence of steps in the design of a Tea or Coffee vending machine, using FSM model only. (08 Marks)
c. Describe the embedded firmware design using ALP with its advantages and disadvantages, along with ALP to MLP conversion procedure. (08 Marks)

OR

- 8 a. Explain any four operational quality attributes in an embedded system design. (04 Marks)
b. Develop and explain the flow in the design of a seat belt warning system, with a sequential program model only. (08 Marks)
c. Describe the embedded firmware design using HLP, with its merits and demerits. How HLP is concreted to MLP in the design process? (08 Marks)

Module-5

- 9 a. Explain the function of a process with a state transition diagram, structure and memory organization of a process. (08 Marks)
b. Explain preemptive SJF scheduling with an example. (06 Marks)
c. Describe deadlock with the various conditions favouring deadlock situation. (06 Marks)

OR

- 10 a. Explain the various units in embedded system development environment with a diagram (IDE). (08 Marks)
b. Explain Round Robin scheduling technique with an example. (06 Marks)
c. Describe the functions of Real Time Kernel in RTOS, in brief. (06 Marks)

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18EC63

Sixth Semester B.E. Degree Examination, Dec.2024/Jan.2025

Microwave and Antenna

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the operation of Reflex Klystron with the help of neat sketch. (10 Marks)
- b. Derive the general transmission line equation to find voltage and current on the line in terms of Position 'Z' and time 't'. (10 Marks)

OR

- 2 a. Explain the mechanism of oscillation of reflex Klystron. (08 Marks)
- b. Define reflection coefficient. Derive the equation for reflection coefficient at the load. (06 Marks)
- c. A transmission line has a characteristic impedance of $50 + j0.01\Omega$ and terminated in a load impedance of $73 - j42.5\Omega$. Calculate :
 - i) Reflection coefficient
 - ii) SWR
 - iii) Transmission coefficient. (06 Marks)

Module-2

- 3 a. Mention the other name of Magic Tee. Derive S matrix of Magic Tee. (08 Marks)
- b. With neat diagram, explain the working of precision type variable attenuator. (06 Marks)
- c. Discuss the following properties of S parameters i) Symmetry of [S] for a reciprocal network. (06 Marks)

OR

- 4 a. Derive the S matrix of E – plane. (06 Marks)
- b. With a neat diagram, explain the working of precision rotary phase shifter. (10 Marks)
- c. Derive S matrix for multiport network. (04 Marks)

Module-3

- 5 a. Derive the characteristic impedance of microstrip line. (06 Marks)
- b. Define the following terms with respect to antenna :
 - i) Beam area
 - ii) Radiation intensity
 - iii) Beam efficiency
 - iv) Directivity
 - v) Radiation resistance. (10 Marks)
- c. A certain microstrip has following parameters :
 $\epsilon_r = 5.23$, $h = 7$ mils, $t = 2.8$ mils, $w = 10$ mils. Calculate the characteristic impedance Z_0 of the line. (04 Marks)

OR

- 6 a. Explain coplanar strip line and shielded strip line. (10 Marks)
 b. A coplanar strip line carries an average power of 250 MW and a peak current of 100mA. Determine the characteristic impedance of the coplanar strip line. (04 Marks)
 c. Define effective aperture. Obtain the relationship between directivity and effective aperture. (06 Marks)

Module-4

- 7 a. Obtain the field pattern for two point source situated symmetrically with respect to the origin. Two sources are fed with equal amplitude and equal phase signals. Assume the distance between two sources $d = \lambda/2$. (06 Marks)
 b. A source has radiation intensity power pattern given by :
 i) $U = U_m \sin^2 \theta$ $0 < \theta < \pi$
 $0 < \phi < 2\pi$
 ii) $U = U_m \cos^2 \theta$ $0 < \theta < \pi/2$
 $0 < \phi < 2\pi$. (08 Marks)
 c. Derive an expression for radiation resistance of a short electric dipole. (06 Marks)

OR

- 8 a. Obtain field expression of two isotropic point source of same amplitude but opposite phase. (06 Marks)
 b. Explain the principle of pattern multiplication with an example. (08 Marks)
 c. Prove that directivity for a source with unidirectional pattern $U_m \cos^n \theta$ where 'n' can be any number, expressed as $D = 2(n + 1)$. (06 Marks)

Module-5

- 9 a. Obtain the expression for radiation resistance of small loop antenna. (08 Marks)
 b. Explain Yagi – Uda array with help of diagram. (06 Marks)
 c. Calculate BWFN, HPBW, directivity and power gain of a uniformly illuminate circular aperture of diameter 8λ . (06 Marks)

OR

- 10 a. Draw the structure of a pyramidal horn antenna. Use the principle of equality of path length and bring out the optimum horn dimension. (06 Marks)
 b. Determine the length L, H plane aperture and flare angles θ_E and θ_H of a pyramidal horn for which E-plane aperture $a_1 = 10\lambda$. The horn is fed by a rectangular wave guide with TE_{10} mode. Let $\delta = 0.2\lambda$ in E-plane and 0.375λ in the H-plane.
 i) What are the beam widths
 ii) What is directivity? (08 Marks)
 c. Calculate BWFN, HPBW, directivity and power gain of a uniformly illuminated circular aperture of diameter 8λ . (06 Marks)

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Sixth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Artificial Neural Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Describe Neuron Activations Correlating to the artificial neuron model with necessary equations. (10 Marks)
- b. Describe the operational procedure of perceptron learning algorithm in detail. (10 Marks)

OR

- 2 a. Apply the weight vector hyper plane orthogonality idea to generate a solution to the NAND problem graphically. (10 Marks)
- b. Explain Binary neurons as pattern dichotomizers. (10 Marks)

Module-2

- 3 a. Illustrate AND pattern classification with detail explanation. (10 Marks)
- b. Discuss weight update procedure in steepest descent search algorithm. (10 Marks)

OR

- 4 a. Derive and explain Back-propagation-algorithm with necessary notations. (10 Marks)
- b. Design a perceptron geometrically to separate the classes as shown in Fig. Q4 (b).

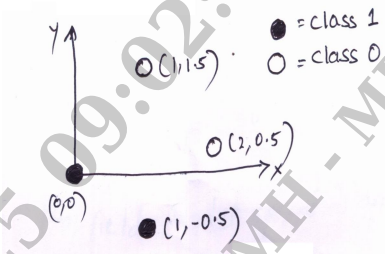


Fig. Q4 (b)

(10 Marks)

Module-3

- 5 a. Apply VC dimension to define the possible ways of linear oriented decision boundary for three points in two-dimensional space. (10 Marks)
- b. Analyze soft margin hyper plane classifier for non-linearly separable data. (10 Marks)

OR

- 6 a. Construct support vector machine for linear separable data and provide complete implications. (10 Marks)
- b. Describe Radial Basis functions with necessary analysis in terms of regularization and generalization. (10 Marks)

Module-4

- 7 a. Derive Hopfield model operation and stability analysis in Discrete time. (10 Marks)
- b. Illustrate additive dynamics in the form electronic circuit interpretation. (10 Marks)

OR

- 8 a. Explain the architecture and operation of Bidirectional Associative memory. (10 Marks)
b. Explain the concept of simulated annealing in detail with all the necessary arguments. (10 Marks)

Module-5

- 9 a. Discuss operational summary of Growing Neural Gas Algorithm. (10 Marks)
b. Explain operational details of the SOFM algorithm. (10 Marks)

OR

- 10 a. Explain the concept of unsupervised vector quantization. (10 Marks)
b. Explain concept of dimensionality reduction using principal component analysis. (10 Marks)

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18EC643

Sixth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Data Structure using C++

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain with example pillars of C++ programming. (10 Marks)
b. Explain the complete structure of C++ programming and write a C++ program to swap 2 numbers. (10 Marks)

OR

- 2 a. Explain the concept of constructor and destructor with simple example. (10 Marks)
b. Explain recursion with simple example. (05 Marks)
c. Explain briefly the concept of polymorphism. (05 Marks)

Module-2

- 3 a. Define array. Also explain the concept of row-major and column – major mapping array. (08 Marks)
b. Explain dynamic memory allocation with example. (06 Marks)
c. With a C++ program to add 2 matrices. (06 Marks)

OR

- 4 a. Explain with example of abstract data type linear list and abstract class linear list. (10 Marks)
b. Explain with diagram singly linked lists and chains. (10 Marks)

Module-3

- 5 a. Explain stack and its applications. (04 Marks)
b. Briefly explain array representation of stack with example. (08 Marks)
c. Explain linked representation of stack. (08 Marks)

OR

- 6 a. Define problem description of tower of Hanoi and write a program to explain the concept of tower of Hanoi using stack. (10 Marks)
b. Write a C++ code to convert infix expression to postfix expression. (10 Marks)

Module-4

- 7 a. Explain ADT specification of queue. (06 Marks)
b. Write C++ program for hash table to perform search and insert operation. (10 Marks)
c. Write a note on class ArrayQueue. (04 Marks)

OR

- 8 a. Define dictionary. Explain operation to be performed on dictionaries. (07 Marks)
b. Write a note on linear list representation of dictionaries. (05 Marks)
c. Briefly explain different collision revaluation technique in hashing. (08 Marks)

Module-5

- 9 a. Explain all properties of binary tree and array representation of binary tree. (07 Marks)
b. Write a C++ program to search a binary search tree. (08 Marks)
c. Write a note on binary tree traversal operation. (05 Marks)

OR

- 10 a. Explain with operation of ADT binary tree. (07 Marks)
b. Write a C++ program to determine height of binary tree. (06 Marks)
c. Describe heap technique and explain heap sorting with relevant diagram. (07 Marks)

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18MAT41

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Complex Analysis, Probability and Statistical Methods

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Show that $w = f(z) = z + e^z$ is analytic and hence find $\frac{dw}{dz}$. (06 Marks)
- b. Derive Cauchy's – Riemann equations in polar form. (07 Marks)
- c. If $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ then find analytic function $f(z) = u + iv$. (07 Marks)

OR

- 2 a. Show that the real and imaginary parts of an analytic function $f(z) = u + iv$ are harmonic. (06 Marks)
- b. If $f(z)$ is an analytic function then show that

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right] |f(z)|^2 = 4 |f'(z)|^2$$
 (07 Marks)
- c. If $u = \left(r + \frac{1}{r} \right) \cos \theta$ then find the corresponding analytic function $f(z) = u + iv$. (07 Marks)

Module-2

- 3 a. State and prove Cauchy's integral formula. (06 Marks)
- b. Discuss the conformal transformation $w = f(z) = z^2$. (07 Marks)
- c. Find the bilinear transformation which maps the points $z = 1, i, -1$ into the points $w = i, 0, -i$. (07 Marks)

OR

- 4 a. Evaluate $\int_C z^2 dz$ along the curve made up of two line segments, one from $z = 0$ to $z = 3$ and another from $z = 3$ to $z = 3 + i$. (06 Marks)
- b. Evaluate $\int_C \frac{e^{2z}}{(z+1)(z-2)} dz$, where C is the circle $|z| = 3$. (07 Marks)
- c. Find the bilinear transformation which maps the points $z = -1, 0, 1$ into the points $w = 0, i, 3i$. (07 Marks)

Module-3

- 5 a. The probability distribution of a random variable X is given by the following table:

$X(=x_i)$	-3	-2	-1	0	1	2	3
$P(X)$	k	2k	3k	4k	3k	2k	k

Find (i) The value of k , (ii) $P(x \leq 1)$, (iii) $P(-1 < x \leq 2)$

(06 Marks)

- b. The probability that a pen manufactured by a factory be defective is $1/10$. If 12 such pens are manufactured, what is the probability that (i) Exactly 2 are defective (ii) Atleast 2 are defective (iii) None of them are defective. (07 Marks)
- c. The length of telephone conversation in a booth has been an exponential distribution and found on an average to be 5 minutes. Find the probability that a random call made from this booth (i) Ends less than 5 minutes (ii) Between 5 and 10 minutes. (07 Marks)

OR

- 6 a. The probability density function of a random variable X is

$$f(x) = \begin{cases} Kx^2 & , \quad 0 < x < 3 \\ 0 & , \quad \text{otherwise} \end{cases}$$

Find (i) The value of K (ii) $P(1 < x < 2)$ (iii) $P(x \leq 1)$ (06 Marks)

- b. In a certain town the duration of a shower is exponentially distributed with mean 5 minutes what is the probability that a shower will last for (i) Ten minutes or more (ii) Less than Ten minutes (iii) Between 10 and 12 minutes. (07 Marks)
- c. The marks of 1000 students in an examination follows a normal distribution with mean 70 and standard deviation 5. Find the number of students whose marks will be (i) less than 65, (ii) more than 75 (iii) 65 to 75. [$\phi(1) = 0.3413$] (07 Marks)

Module-4

- 7 a. Compute the rank correlation coefficient for the following data:

x	68	64	75	50	64	80	75	40	55	64
y	62	58	68	45	81	60	68	48	50	70

(06 Marks)

- b. Find a best fitting straight line $y = ax + b$ for the data below:

x	1	3	4	6	8	9	11	14
y	1	2	4	4	5	7	8	9

(07 Marks)

- c. Obtain the lines of regression and hence find the coefficient of correlation for the data below:

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

(07 Marks)

OR

- 8 a. If θ is the acute angle between the lines of regression then show that

$$\tan \theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left[\frac{1 - r^2}{r} \right] \quad (06 \text{ Marks})$$

- b. Find a best fitting second degree parabola of the form $y = ax^2 + bx + c$ for the data below:

x	1	2	3	4	5
y	10	12	13	16	19

(07 Marks)

- c. Find the coefficient of correlation for the following data:

x	10	14	18	22	26	30
y	18	12	24	06	30	36

(07 Marks)

Module-5

- 9 a. The joint probability of discrete random variables X and Y is given below:

Y \ X	1	3	9
2	1/8	1/24	1/12
4	1/4	1/4	0
6	1/8	1/24	1/12

Determine (i) Marginal distribution of X and Y. (ii) Covariance and correlation of X and Y.

(06 Marks)

- b. A survey was conducted in a slum locality of 2000 families by selecting a sample of size 800, it was revealed that 180 families were illiterates. Find the probable limits of the illiterates families in the population of 2000 at 1% level of significance. (07 Marks)
- c. A group of 10 boys fed at diet A and another group of 08 boys fed on another diet B for a period of 06 months record the following increase in weights in pounds.

Diet A	05	06	08	01	12	04	03	09	06	10
Diet B	02	03	06	08	10	01	02	08	-	-

Test whether diet A and B differ significantly regarding their effect on increase in weight. [$t_{0.05} = 2.12$]

(07 Marks)

OR

- 10 a. Explain the terms :
 (i) Null hypothesis
 (ii) Type-I and Type-II errors
 (iii) Level of significance. (06 Marks)
- b. A certain stimulus administered to each of the 12 patients resulted in the following change in blood pressure: 5, 2, 8, -1, 3, 0, 6, -2, 1, 5, 0, 4
 Can it be concluded that the stimulus will increase the blood pressure? [$t_{0.05} = 2.201$] (07 Marks)
- c. A sample analysis of examination, result of 500 students was made, it was found that 220 students had failed, 170 had secured third class, 90 had secured second class, 20 had secured first class. Do these figures support the general examination result which is in the ratio 4 : 3 : 2 : 1 for the respective categories? [$\chi^2_{0.05} = 7.81$]. (07 Marks)

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